



Data Model Design Document

End-to-End Salesforce CRM Transformation for
Customer Operations and Service Management

4.1 Object Definition

To establish a scalable architecture that accurately represents physical infrastructure assets while preventing duplicate records, a balanced combination of standard and custom objects has been structured below.

4.1.1 Standard Objects

- **Account:** Represents the primary consumer entity or household receiving active utility distribution services from GreenGrid.
- **Contact:** Manages the detailed personal identification information, phone numbers, and communication logs for individual billing representatives.
- **Case:** Serves as the central transactional ticket to log, prioritize, track, and resolve active customer service telephone complaints.

4.1.2 Custom Objects

- **Utility_Asset__c:** Tracks the physical utility meter hardware grid installation asset. This object stores the alphanumeric identifier field natively within Salesforce to guarantee high-performance search metrics.
- **Field_Visit__c:** Tracks the history of remote on-site dispatches, travel variables, repair assignments, and technical resolution details handled by the field crew.

4.2 Field Definition

4.2.1 Standard Object

To accommodate the specialized customer service workflows needed for utility infrastructure management, the standard Salesforce Case object was customized. Instead of introducing a redundant custom entity, the out-of-the-box **Type** field metadata was expanded with industry-specific picklist values. This classifies incoming support tickets efficiently while leveraging standard platform optimization.

Standard Object: Case

Field Label	API Name	Data Type	Description and Validation Rules
Case Type	Type	Picklist	* Outage: Used to classify distribution grid failures. * Maintenance: Used for scheduling routine technical grid adjustments and asset tracking updates to mitigate active failure risks.

4.2.2 Custom Object

The tables below outline the critical technical fields needed to execute business logic, enforce data validation, and meet the requirement of under 2-second search performance metrics.

Custom Object: Utility_Asset__c

Field Label	API Name	Data Type	Description and Validation Rules
Asset Name	Name	Auto-Number	Automatically generated system identifier (Format: MET-{0000}).

Meter ID	Meter_ID__c	Text (Length: 50)	Unique, indexed identifier field used by agents for rapid customer lookups. Set as Unique and External ID .
Asset Status	Asset_Status__c	Picklist	Tracks operational state. Values: <i>Active</i> , <i>Inactive</i> , <i>Suspended</i> .
Installation Address	Installation_Address__c	Text Area (Length: 255)	Holds the exact physical geographic address where the meter asset is deployed.
Customer Account	Customer_Account__c	Lookup (Account)	Establishes a flexible relationship linking the physical hardware asset to a parent customer account.

Custom Object: Field_Visit__c

Field Label	API Name	Data Type	Description and Validation Rules
Visit Number	Name	Auto-Number	Automatically generated record identifier (Format: VIS-{0000}).
Parent Case	Parent_Case__c	Lookup (Case)	Connects the dispatch log to its original support ticket. <i>Validation Rule:</i> Clear value on delete or restrict deletion to preserve history.
Repair Status	Repair_Status__c	Picklist	Mandatory operational tracking field. Values: <i>Assigned</i> , <i>In Progress</i> , <i>Resolved</i> , <i>Failed</i> .

Technician Notes	Technician_Notes__c	Long Text Area (32,000)	Long text field where technicians type site findings directly into the mobile layout.
Resolution Timestamp	Resolution_Timestamp__c	Date/Time	Automatically populated by system flows when status shifts to <i>Resolved</i> to track KPI dashboards.

4.3 Relationship Mapping

This section details the relational structure of the database schema, illustrating how standard and custom entities link together within the GreenGrid Utilities Salesforce environment. By applying specific database constraints, this architecture protects regulatory historical compliance data and enforces reference integrity.

The matrix below defines the relationship types, parent-child structures, and exact transactional system behaviors applied during deletion scenarios.

Parent Object	Child Object	Relationship Type	Data Behavior and Deletion Constraint
Account	Contact	Standard Lookup	Clear Value on Delete: If an Account record is deleted, related Contact records remain in the database with their lookup reference field cleared.
Account	Utility_Asset__c	Custom Lookup	Clear Value on Change: If a customer account is modified or closed, the physical meter asset record remains un-deleted and can be re-assigned to the incoming occupant account.

Account	Case	Standard Lookup	Restricted Deletion: The system blocks the deletion of an Account record if active or historical Case records are attached to it, securing critical customer audit logs.
Utility_Asset__c	Case	Custom Lookup	Clear Value on Change: If hardware assets are replaced or updated, historical service ticket logs remain safely intact for utility grid pattern analytics.
Case	Field_Visit__c	Custom Lookup	Restricted Deletion: The system prevents the manual deletion of an active Field Visit record if it contains unresolved technical notes, ensuring audit compliance for field operations.

- **Account to Contact Relationship:** Formed as a standard Lookup connection. This allows a primary customer household account to act as the main hub for multiple individual contact records (such as secondary residents or property managers) while keeping data independent.
- **Account to Utility_Asset__c Relationship:** Designed as a custom Lookup connection. This choice uncouples the physical utility meter hardware from changing human residents occupying the property. When property ownership changes, the asset record remains bound to its geographic installation address, and the lookup field is simply updated to the new Account.
- **Account to Case Relationship:** Configured as a standard Lookup connection governed by a strict Restricted Deletion rule. This safeguard ensures that legacy support interactions, outage history, and resolution timelines remain archived for corporate SLA reporting, even if a user tries to delete the primary customer account file.

- **Utility_Asset__c to Case Relationship:** Set up as a custom Lookup connection. Linking the ticket directly to the hardware asset allows helpdesk agents to instantly check the asset's mechanical repair history during an active call, speeding up root-cause diagnosis.
- **Case to Field_Visit__c Relationship:** Set up as a custom Master-Detail connection. This tight structural bond ensures that field workers can only generate a field visit if it is tied to an active, authorized parent case. Additionally, the child record automatically inherits the security visibility rules, owner assignments, and sharing permissions defined on the parent Case.

4.4 Salesforce Implementation and Configuration

Step 1: Creating a Developer Account

1. Navigate to the Salesforce Developer Edition Sign-up page.
2. Fill out the registration form with your details (First Name, Last Name, Email, Role, Company, Country, and Postal Code).

salesforce.com/products/free-trial/developer/

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Job title: Student ✓ Company: SmartBridge ✓

Country/region: Indonesia ✓ Work email: syakilaaini61@gmail.com ✓

☒ I agree to the [Main Services Agreement - Developer Services and Salesforce Program](#)

☒ Yes, I would like to receive marketing communications regarding Salesforce products, services, and events. I can unsubscribe at any time.

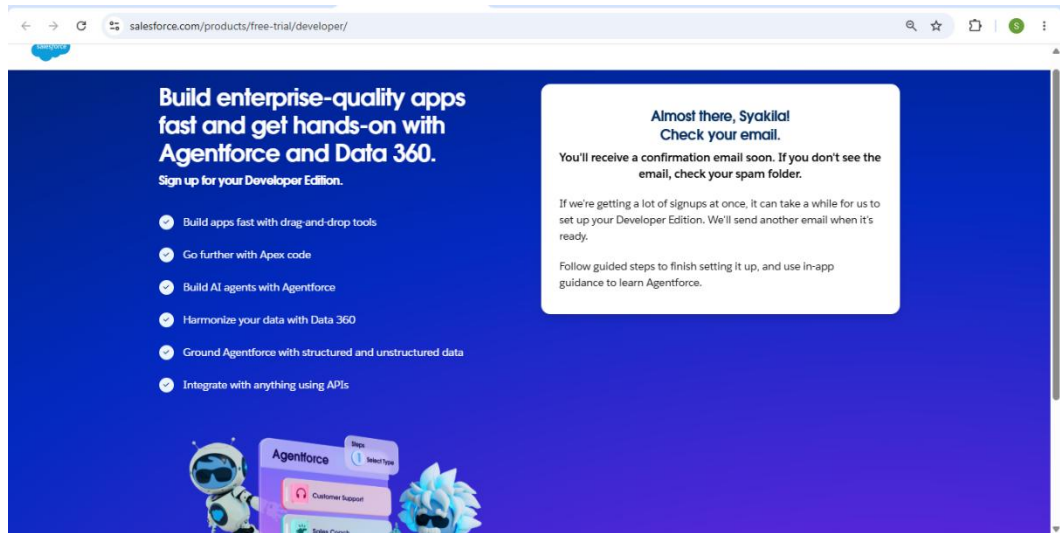
Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features, and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

By submitting this form, you confirm that you agree to the processing of your personal data, including transfer and processing outside of Indonesia, as described in the [privacy statement](#).

[Sign me up](#)

3. Click the **"Sign me up"** button and wait for the confirmation email to arrive in your inbox.



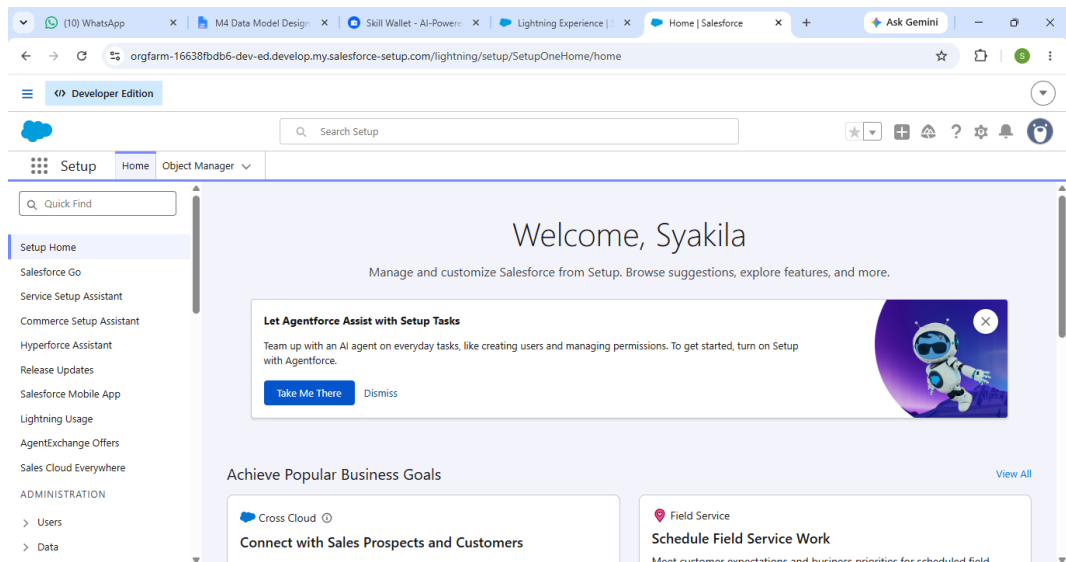
Step 2: Account Activation

1. Access the inbox of the email address provided during sign-up.
2. Open the verification email and click the **"Reset Password"** button to activate your account (this email may take 2 to 10 minutes to arrive).
3. Set up your new password, choose a security question, provide an answer, and click **"Change Password"**.
4. You will be automatically redirected to the Salesforce Setup home screen.

Step 3: Custom Object Creation for Utility Asset

The primary hardware tracking table was established by generating a new custom object inside the Object Manager tool. Follow these actions:

1. From the Setup home page, click on the **Object Manager** tab.
2. Click **Create** in the upper right-hand corner, then select **Custom Object**.



3. Define the object configuration parameters as follows:

- **Label:** Utility Asset
- **Plural Label:** Utility Assets
- **Object Name (API Name):** Utility_Asset
- **Record Name:** Asset Name
- **Data Type:** Auto-Number (Display Format: MET-{0000}, Starting Number: 1)

4. Under **Optional Features**, check the following boxes:

- *Allow Reports*
- *Allow Track Field History*

5. Under **Search Status**, check **Allow Search**.

6. Click **Save**.

Custom Object Definition Edit

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label: Example: Account

Plural Label: Example: Accounts

Starts with vowel sound: ☐

The Object Name is used when referencing the object via the API.

Object Name: Example: Account

Description:

Context-sensitive help setting: ☒ Open the standard Salesforce.com Help & Training window ☐ Open a window using a Visualforce page

Context name:

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name: Example: Account Name

Data Type: Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Display Format: Example: A-00000 What Is This?

Starting Number:

Optional Features

☒ Allow Reports ☐ Allow Activities ☒ Track Field History ☐ Allow in Chatter Groups ☐ Enable Licensing

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

☒ Allow Sharing ☒ Allow Bulk API Access ☒ Allow Streaming API Access

Deployment Status

☐ In Development ☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

☒ Allow Search

Object Creation Options (Available only when custom object is first created)

☐ Add Notes and Attachments related list to default page layout ☐ Launch New Custom Tab Wizard after saving this custom object

Step 4: Custom Field Creation for Utility Asset

Specific technical metadata fields were configured under the **Fields & Relationships** section of the new Utility Asset object to capture operational data:

Fields & Relationships

8 Items, Sorted by Field Label

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Asset Name	Name	Auto Number		✓
Asset Status	Asset_Status__c	Picklist		
Created By	CreatedById	Lookup(User)		
Customer Account	Customer_Account__c	Lookup(Account)		✓
Installation Address	Installation_Address__c	Text Area(255)		
Last Modified By	LastModifiedById	Lookup(User)		
Meter ID	Meter_ID__c	Text(50)		
Owner	OwnerId	Lookup(User/Group)		✓

- **Meter ID (Meter_ID__c):** Created as a Text field with a character length of 50. To satisfy the high-priority business requirement of sub-2-second data retrieval speeds, this attribute was explicitly enabled as both a **Unique** record identifier and an **Indexed External ID**.
- **Asset Status (Asset_Status__c):** Configured as a Picklist data type containing explicit system values: *Active, Inactive, and Suspended*.
- **Installation Address (Installation_Address__c):** Configured as a standard Text Area field capped at 255 characters to efficiently store geographic location strings.
- **Customer Account (Customer_Account__c):** Established as a flexible Lookup Relationship pointing directly to the standard Account entity to connect assets to active consumers.

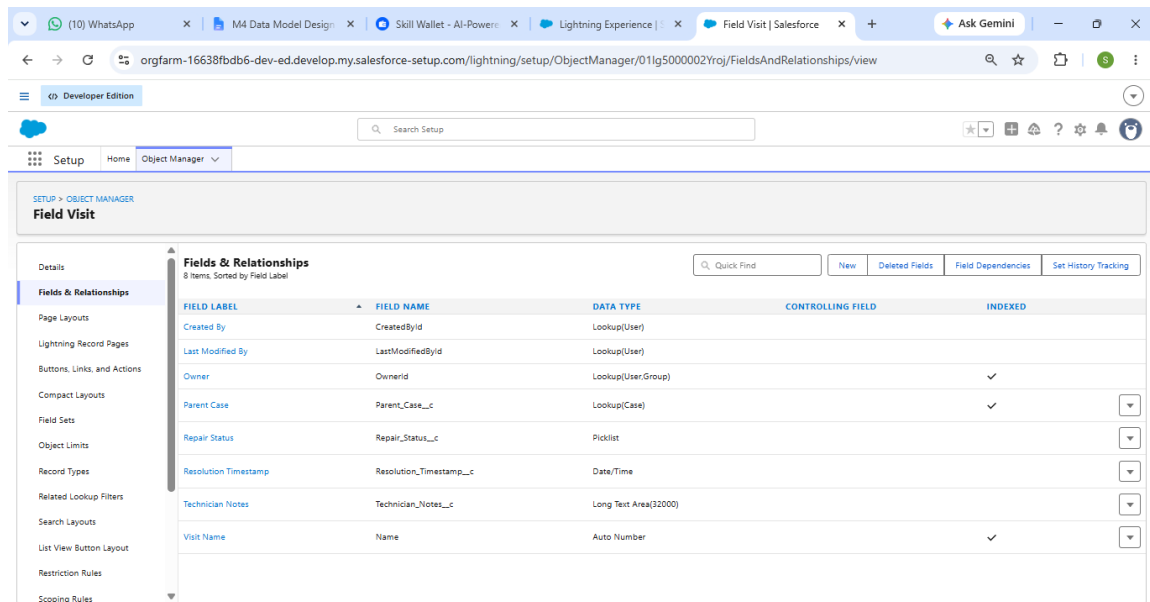
Step 5: Custom Object Creation for Field Visit

To move the technical field crew dispatch processes away from legacy paper-based tracking methods, a secondary custom object was created inside the Object Manager layout.

- **Label:** Field Visit
- **Plural Label:** Field Visits
- **Object Name (API Name):** Field_Visit
- **Record Name:** Visit Number
- **Data Type:** Auto-Number (Display Format: VIS-{0000}, Starting Number: 1)
- *Follow the identical creation steps outlined in Step 3.*

Step 6: Custom Field Creation for Field Visit

To maintain operational tracking flexibility and support offline mobile user requirements, data properties were added to the Field Visit object framework:



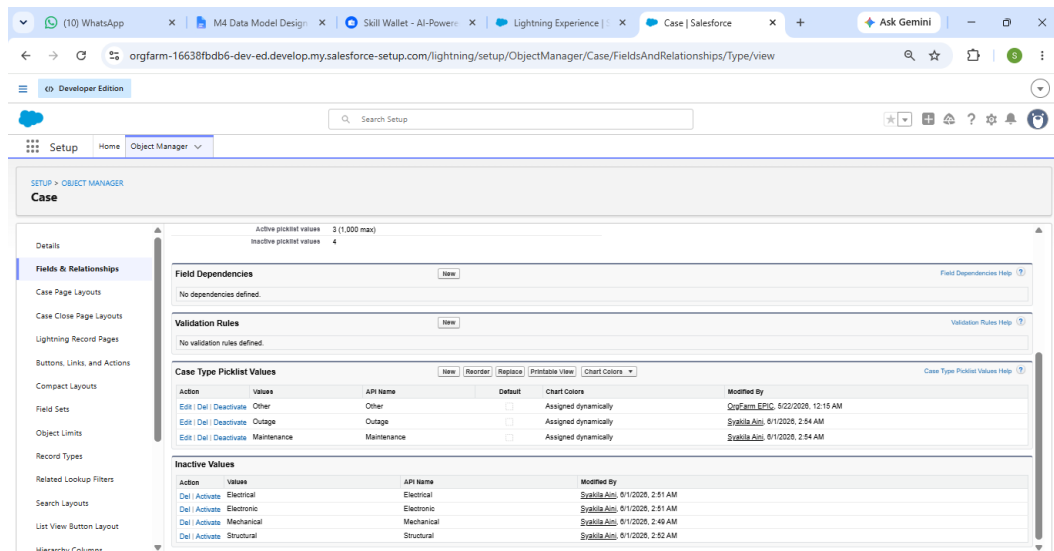
- **Parent Case (Parent_Case__c):** Set up as a standard Lookup Relationship pointing directly to the Case entity. This structural choice ensures that field workers can maintain independent record ownership on their mobile devices while staying connected to the root support ticket.
- **Repair Status (Repair_Status__c):** Configured as a Picklist data type with workflow status values: *Assigned, In Progress, Resolved, and Failed*.
- **Technician Notes (Technician_Notes__c):** Configured as a Long Text Area (32,000 characters) to allow detailed mobile technical descriptions.
- **Resolution Timestamp (Resolution_Timestamp__c):** Configured as a Date/Time field to log the exact completion moment for operational efficiency metrics.

Step 7: Standard Object Metadata and Page Layout Customization (Case)

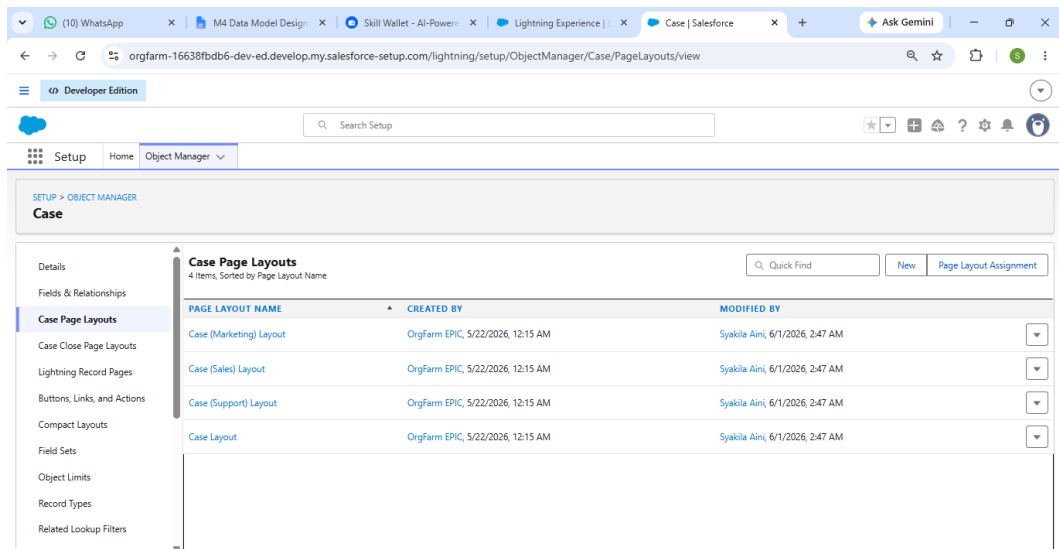
Before verifying the final system schema, the standard Case object infrastructure is customized to recognize utility-specific support categories and clean up the user interface.

1. From the Setup page, click on **Object Manager** in the top menu bar.
2. Scroll down and click on the standard **Case** object.
3. Choose **Fields & Relationships**, then click the **New** button.
4. Select **Lookup Relationship** as the Data Type and click **Next**.

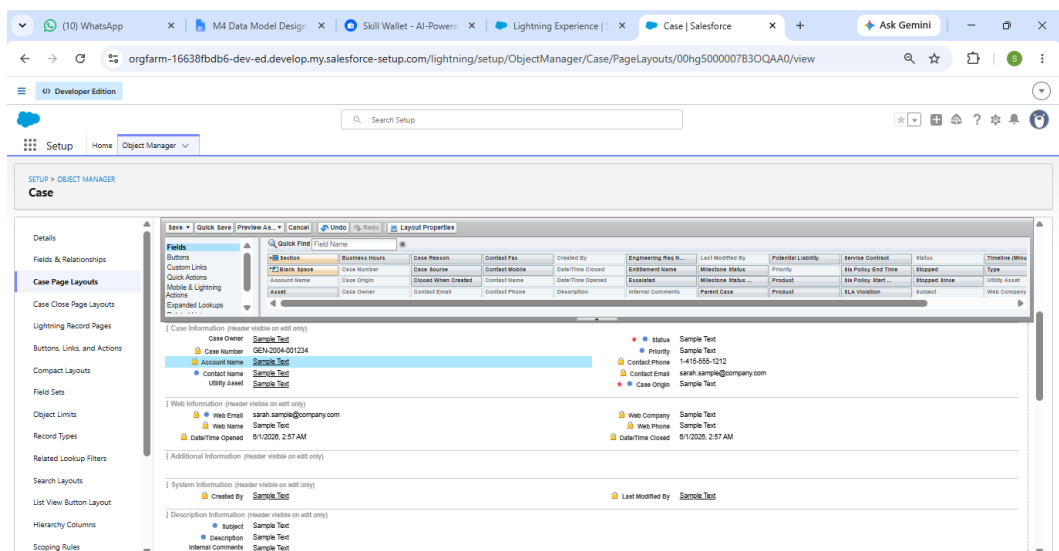
5. In the **Related To** dropdown menu, select **Utility Asset** and click **Next**.
6. Set the Field Label to Utility Asset (the Field Name will auto-populate as Utility_Asset) and click **Next**.
7. Advance through the field-level security visibility screens, leave the field added to the layouts, and click **Save**.
8. Remaining under Case Fields & Relationships, click on the standard field **Type** (Picklist).
9. Scroll down to the **Case Type Picklist Values** related list and click **New** to add Outage and Maintenance on separate lines. Click **Save**.



10. To clean up the interface and enforce strict data standardization, locate the standard default values (*Mechanical, Electrical, Electronic, Structural, and Other*) within the active values table, and click **Del / Deactivate** next to each entry.
11. From the left navigation menu of the Case object, click on **Case Page Layouts** and select **Case Layout**.



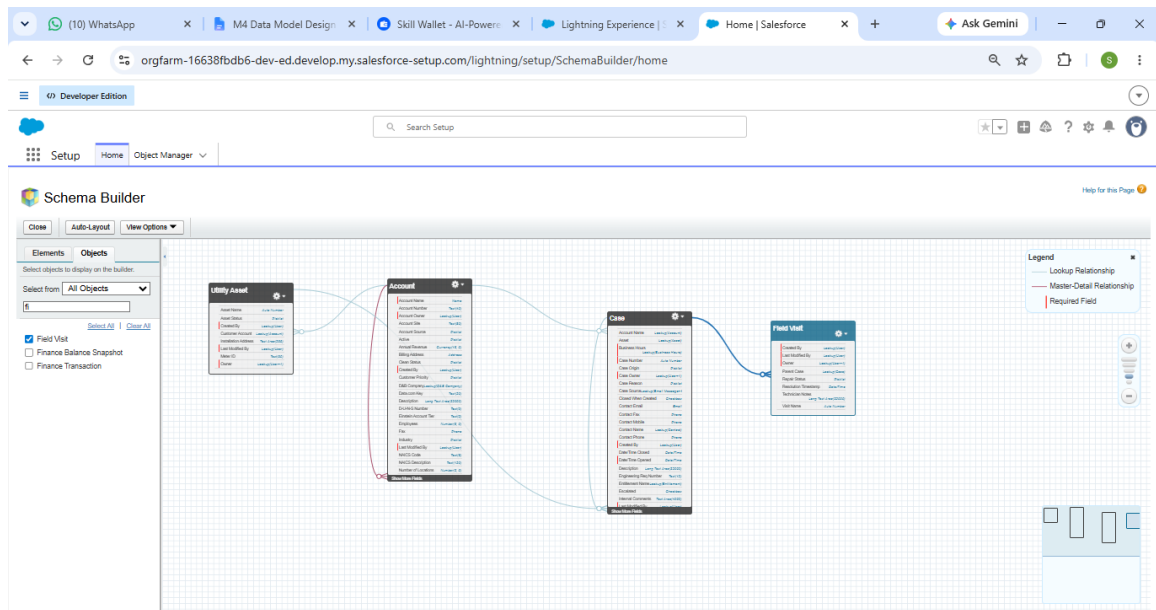
12. Streamline the Case Page Layout by removing irrelevant standard fields and positioning the custom **Utility Asset** and updated **Type** fields prominently.



13. Click **Save** on the upper left-hand layout palette menu.

Step 8: System Schema Integrity Verification

The native Salesforce **Schema Builder** utility was utilized to run a database layout audit. By isolating the standard Account and customized Case entities alongside the custom Utility Asset and Field Visit objects, the data model structure was visually checked to ensure reference integrity before entering the security implementation phase.



Step 9: Custom Lightning App Creation

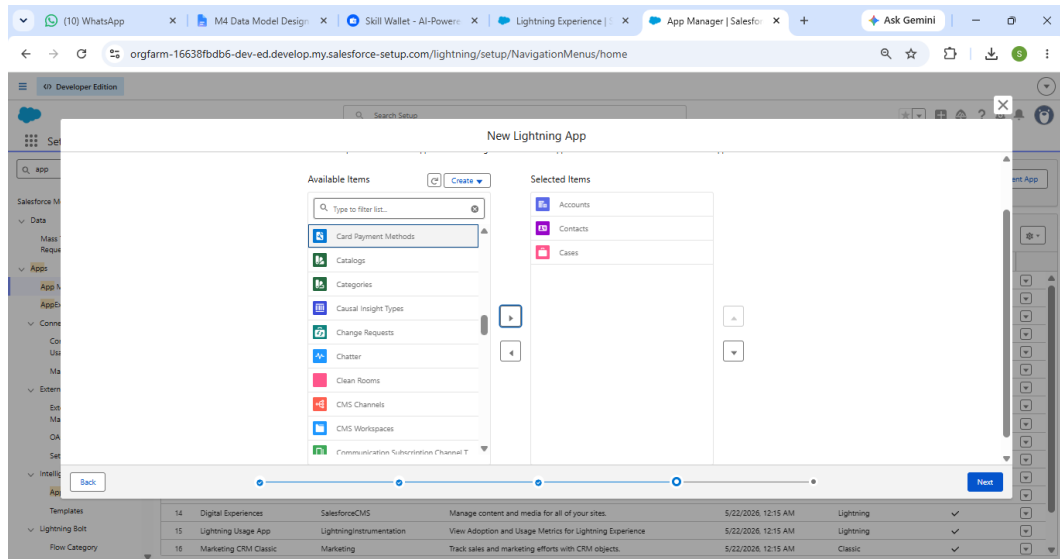
To establish a dedicated, branded workspace layout for end-users, a custom Lightning Application container was provisioned within the platform App Manager.

1. Navigate to **Setup** → type **App Manager** in the Quick Find bar, click on **App Manager**.
2. Click the **New Lightning App** button located in the upper right-hand corner.
3. Fill out the App Name as GreenGrid Utilities, upload branding assets if needed, and click **Next**.

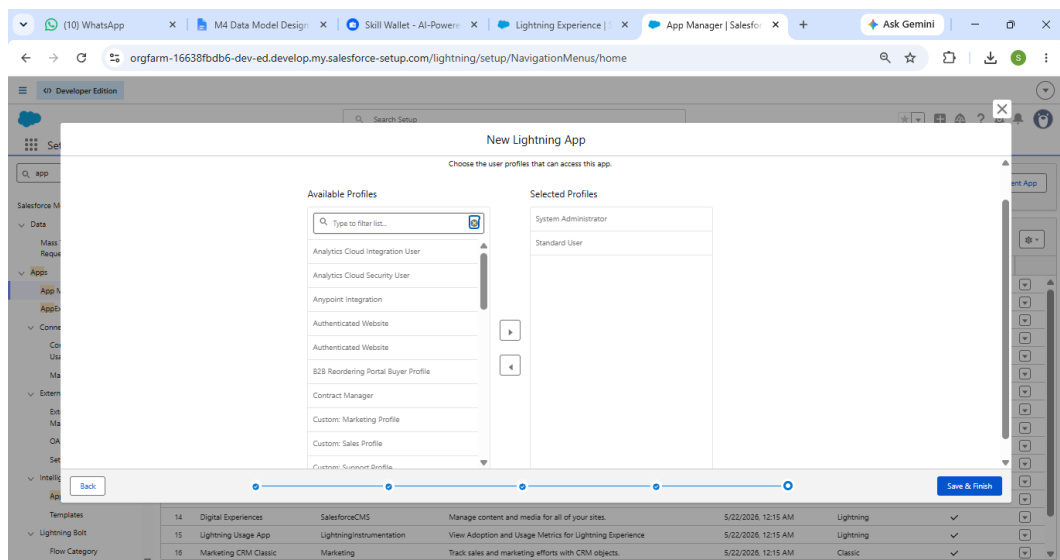
The screenshot shows the 'New Lightning App' form in the Salesforce App Manager. The form is titled 'New Lightning App' and contains two main sections: 'App Details' and 'App Branding'. The 'App Details' section has three fields: 'App Name' (filled with 'GreenGrid Utilities'), 'Developer Name' (filled with 'GreenGrid.Utilities'), and 'Description' (with a placeholder 'Enter a description...'). The 'App Branding' section has an 'Image' field (with a GreenGrid logo), a 'Primary Color Hex' field (with a value of '#0070C2'), and a 'Clear' button. Below these is the 'App Launcher Preview' showing the 'GreenGrid Utilities' app icon. At the bottom right, there is a 'Next' button. The background shows the Salesforce App Manager interface with a sidebar on the left and a table of existing apps at the bottom.

Template	ID	Name	Category	Description	Created	Version	Status
Digital Experiences	14	SalesforceCMS	Manage content and media for all of your sites.	5/22/2026, 12:15 AM	Lightning	✓	
Lightning Usage App	15	LightningInstrumentation	View Adoption and Usage Metrics for Lightning Experience	5/22/2026, 12:15 AM	Lightning	✓	
Marketing CRM Classic	16	Marketing	Track sales and marketing efforts with CRM objects.	5/22/2026, 12:15 AM	Classic	✓	

4. Advance through the **App Options** and **Utility Items** parameters by keeping the standard system default settings.
5. On the **Navigation Items** panel, select standard entities **Accounts**, **Contacts**, and **Cases** from the available list and move them to the **Selected Items** box to form the base configuration baseline.



6. On the **User Profiles** assignment step, scroll down and assign active visibility access to the **System Administrator** profile along with core operational user profiles.

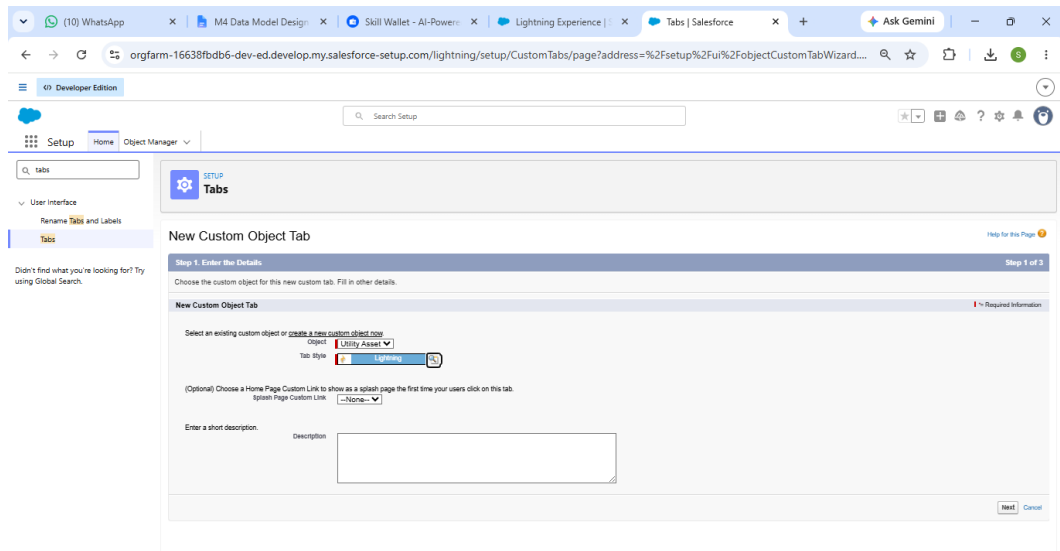


7. Click **Save & Finish**.

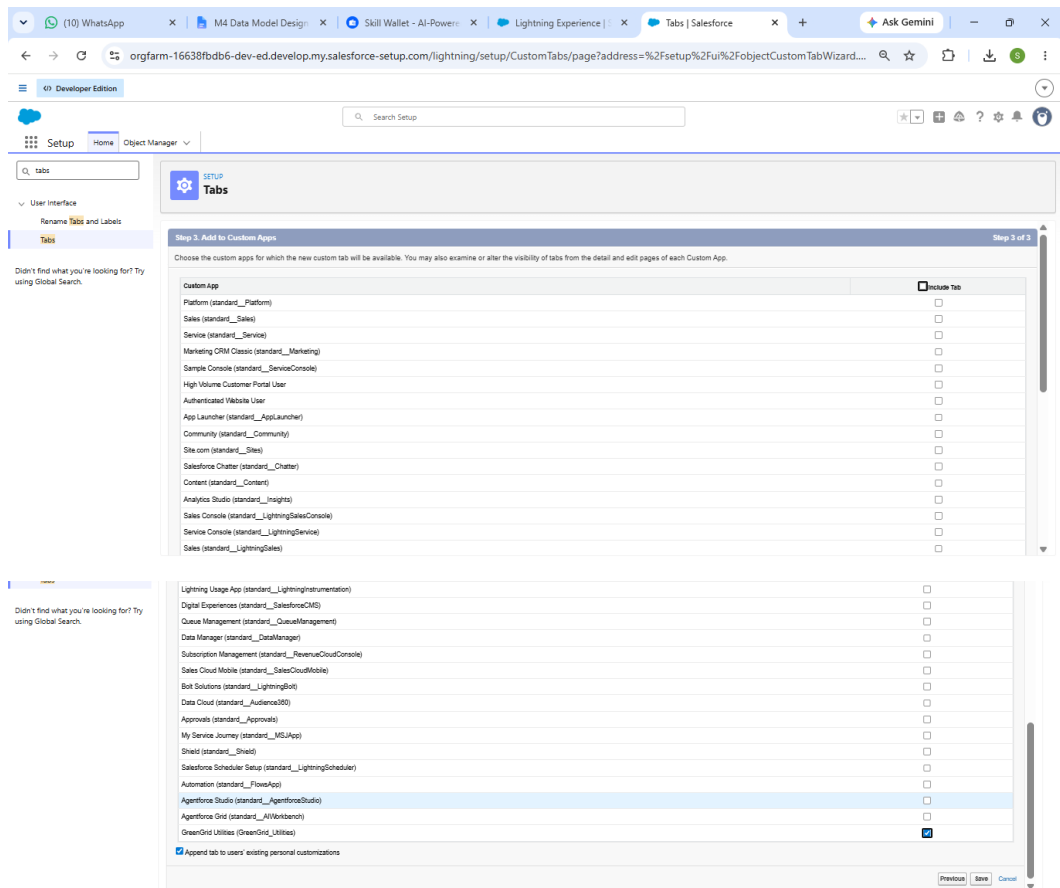
Step 10: Custom Object Tab Provisioning (Utility Asset)

To make the newly deployed custom tables visible and accessible to system end-users within the navigation interface, standalone user interface tabs were provisioned.

1. Navigate to **Setup**, type **Tabs** in the Quick Find bar, click on **Tabs**.
2. Under the **Custom Object Tabs** related list, click the **New** button.



3. Select **Utility Asset** from the Object dropdown list and assign a distinct visual **Tab Style**. Click **Next**.
4. Leave the profile visibility settings at their default values and click **Next** again.
5. On the **Add to Custom Apps** step, uncheck the "Include Tab" global header box to remove it from irrelevant background applications, but ensure the checkbox remains explicitly checked for the **GreenGrid Utilities App**.

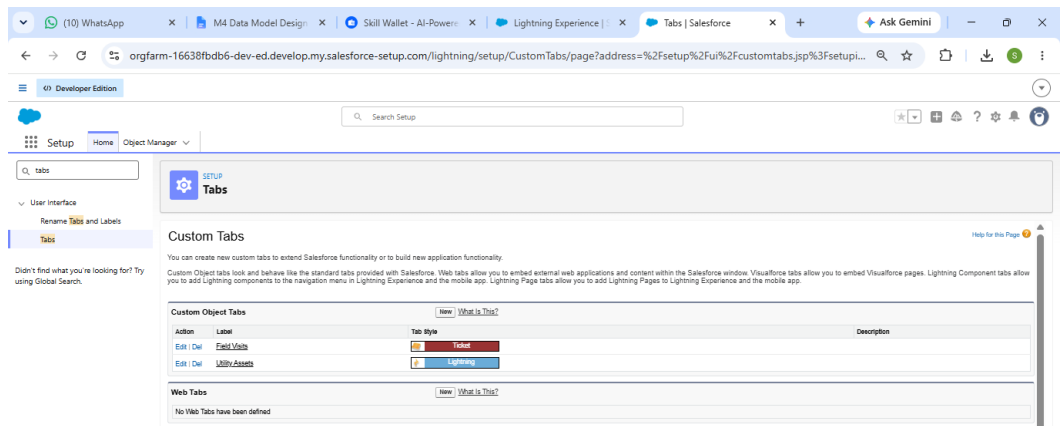


6. Ensure the checkbox for "**Append tab to users' existing personal customizations**" is checked at the bottom of the page layout.
7. Click **Save**.

Step 11: Custom Object Tab Provisioning (Field Visit)

Following the successful deployment of the primary infrastructure layout, navigation tabs for the remaining operational objects were built using the same method as Step 10 to complete the end-user navigation framework.

1. Return to the **Tabs** menu and click **New** under the Custom Object Tabs panel.
2. Select **Field Visit** from the Object dropdown list and assign a distinct visual **Tab Style**.
3. Advance to the app routing parameters screen, exclude the tab from general background profiles, and explicitly enable it for the **GreenGrid Utilities App** navigation bar.



4. Verify the personal customization append rule is active and click **Save**.